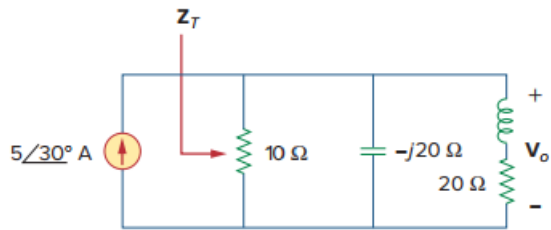


Problem

Find $\mathbf{Z}_T$ and $\mathbf{V}_o$ in the circuit in Fig. 9.71. Let the value of the inductance equal $j20\ \Omega$.

Figure 9.71:



Step-by-step solution

Step 1 of 4

Refer to Figure 9.71 in the text book.

Draw the circuit diagram to determine the equivalent total impedance.

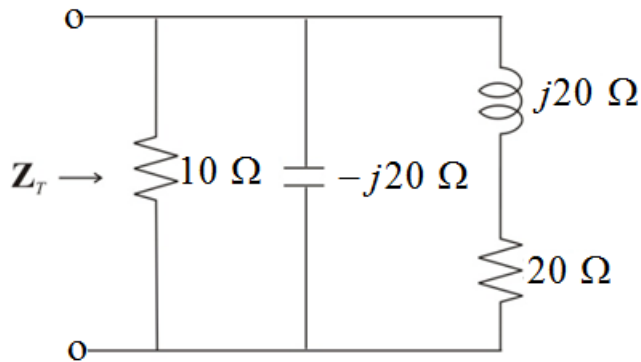


Figure 1

Step 2 of 4

Calculate the value of the impedance $\mathbf{Z}_T$.

$$\begin{aligned}\mathbf{Z}_T &= (10) \parallel (-j20) \parallel (20 + j20) \\ &= (10) \parallel \frac{(-j20)(20 + j20)}{(-j20) + (20 + j20)} \\ &= (10) \parallel \frac{400 - j400}{20} \\ &= (10) \parallel (20 - j20)\end{aligned}$$

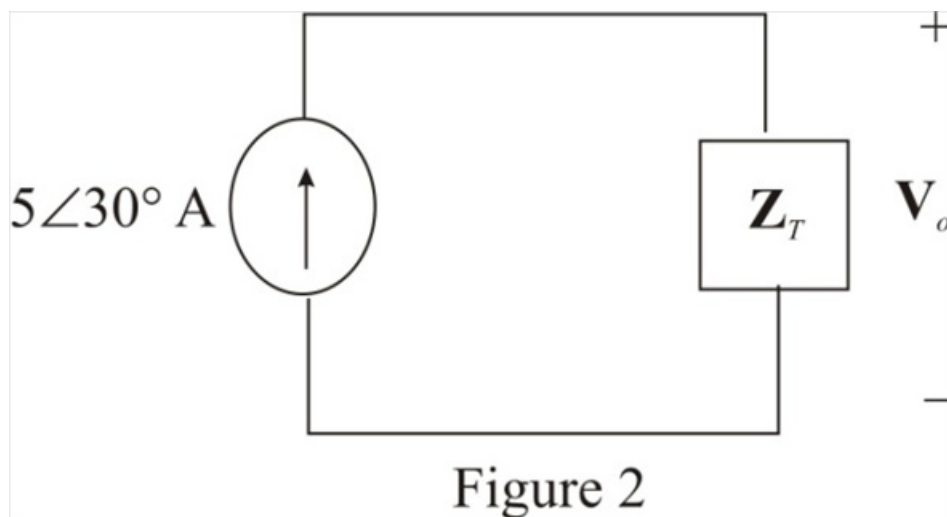
Further simplification as follows,

$$\begin{aligned}\mathbf{Z}_T &= \frac{(10)(20 - j20)}{(10) + (20 - j20)} \\ &= \frac{200 - j200}{30 - j20} \\ &= 7.6923 - j1.5385 \\ &= 7.845 \angle -11.31^\circ \Omega\end{aligned}$$

Thus, the value of the impedance, $\mathbf{Z}_T$ is $7.845 \angle -11.31^\circ \Omega$.

Step 3 of 4

Draw the simplified circuit diagram.



Step 4 of 4

Determine the value of the voltage V_o .

$$V_o = (5\angle 30^\circ)(Z_T)$$

Substitute $7.845\angle -11.31^\circ \Omega$ for Z_T in the equation.

$$\begin{aligned} V_o &= (5\angle 30^\circ)(7.845\angle -11.31^\circ) \\ &= 39.225\angle 18.69^\circ \text{ V} \end{aligned}$$

Thus, the value of the output voltage, V_o is $39.225\angle 18.69^\circ \text{ V}$.